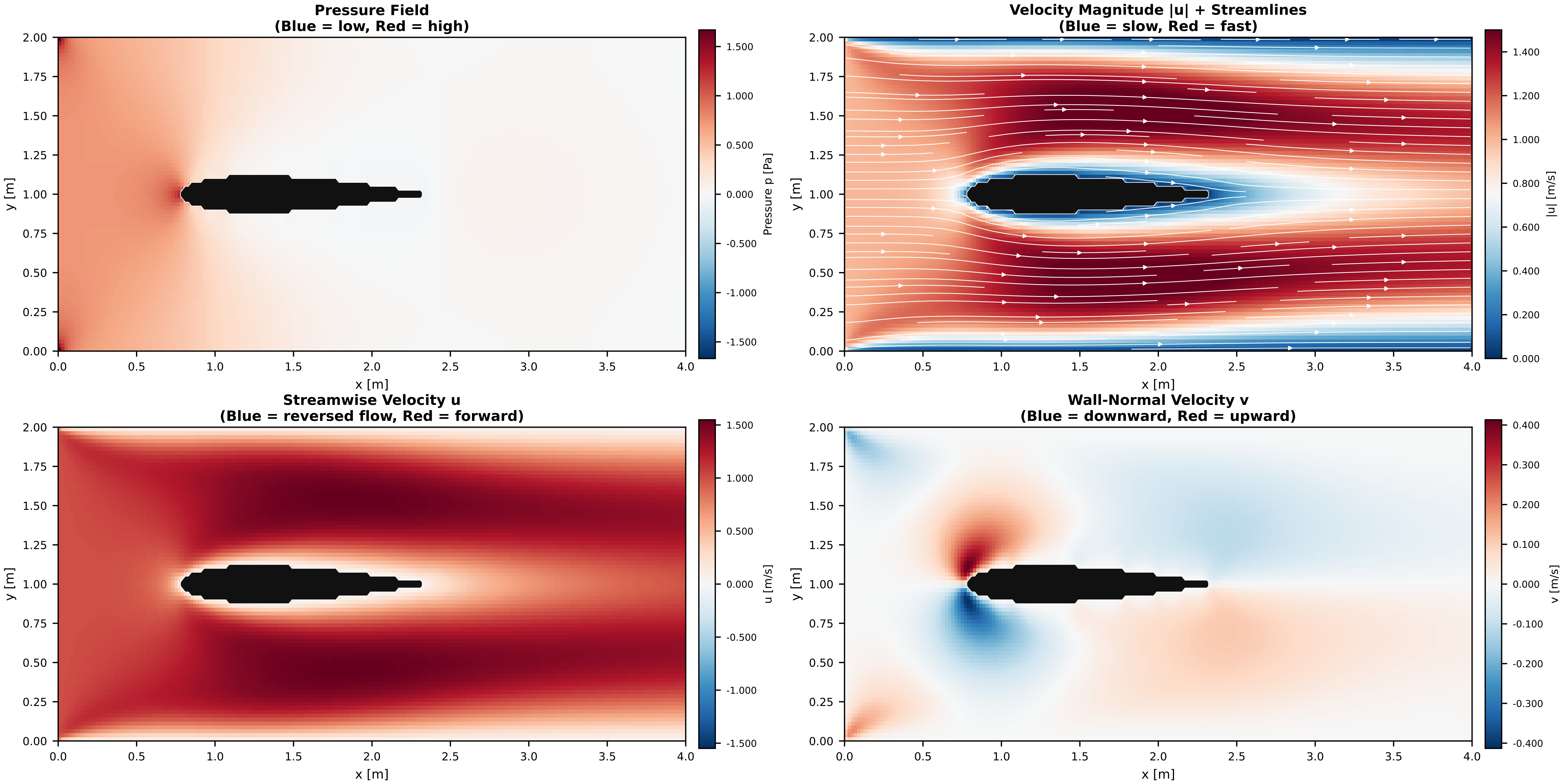
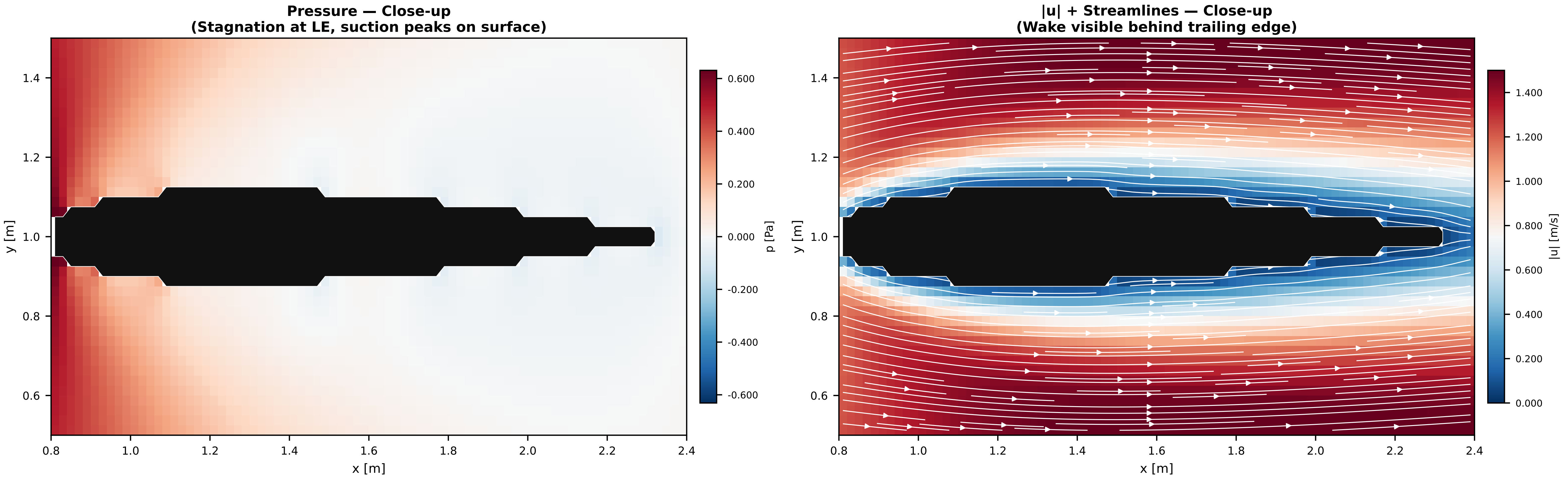
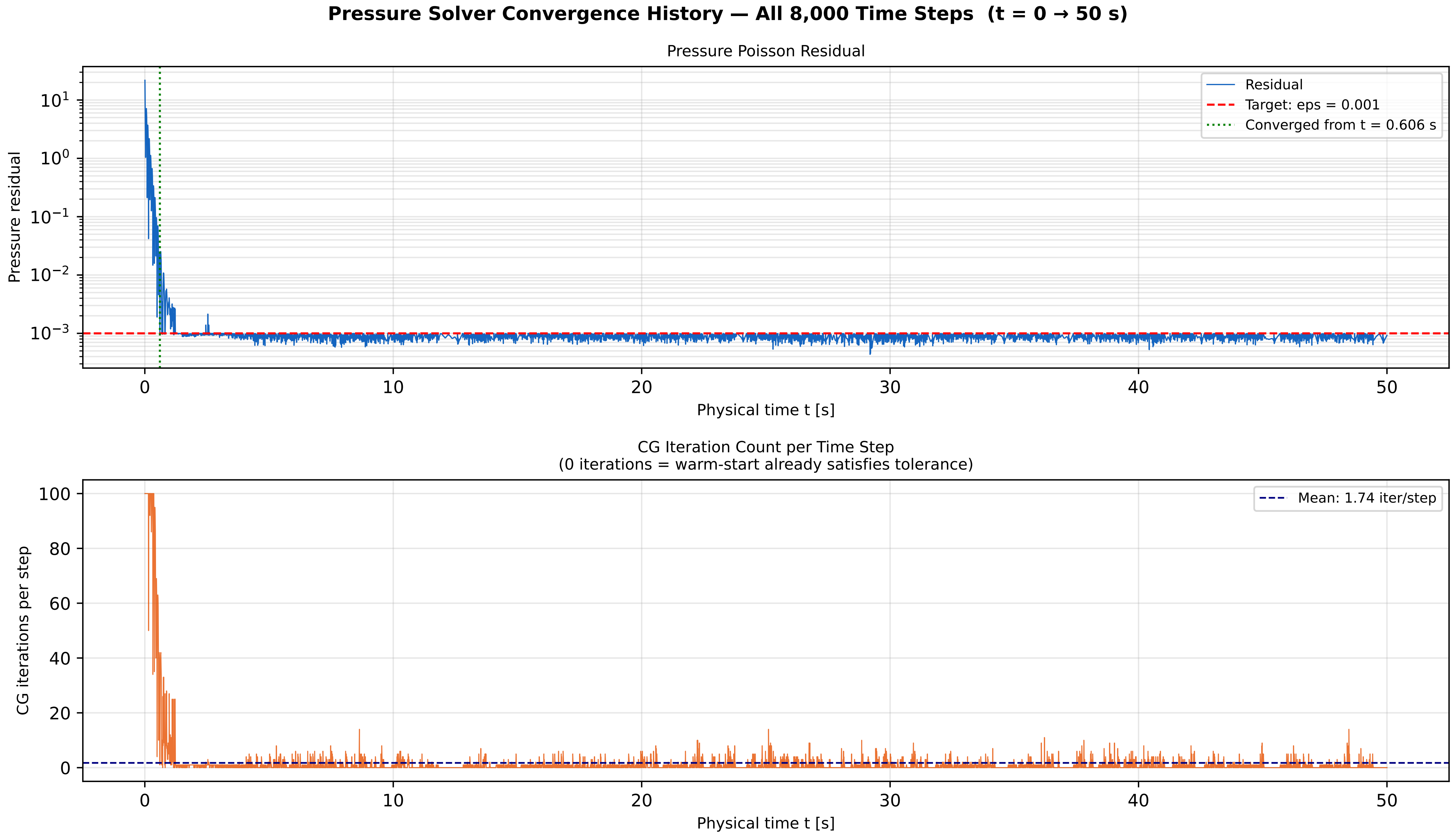
NACA 0012 Airfoil Wind Tunnel — Flow Field Overview

Re = 200, α = 0°, Solver: EIGEN_CG (serial), Grid: 200×80, Steady state at t ≈ 45 s

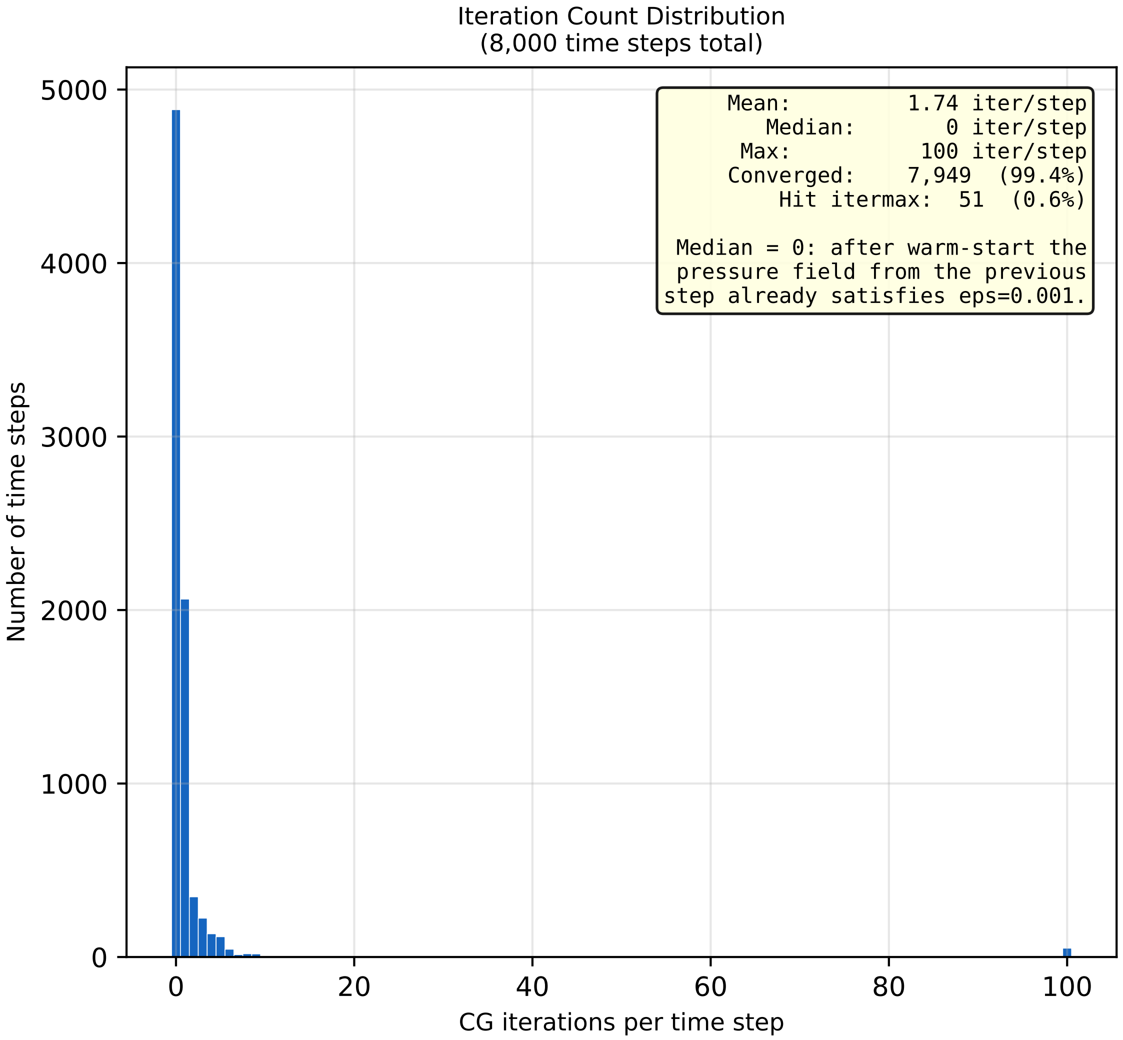


Close-up View — Airfoil Region (x ∈ [0.8, 2.4], y ∈ [0.5, 1.5])




Pressure Solver Convergence History — All 8,000 Time Steps (t = 0 → 50 s)
Pressure Poisson Residual
Residual
Target: eps = 0.001
Converged from t = 0.606 s
Pressure residual
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
Physical time t [s]
0
10
20
30
40
50
CG Iteration Count per Time Step
(0 iterations = warm-start already satisfies tolerance)
Mean: 1.74 iter/step
CG iterations per step
100
80
60
40
20
0
Physical time t [s]

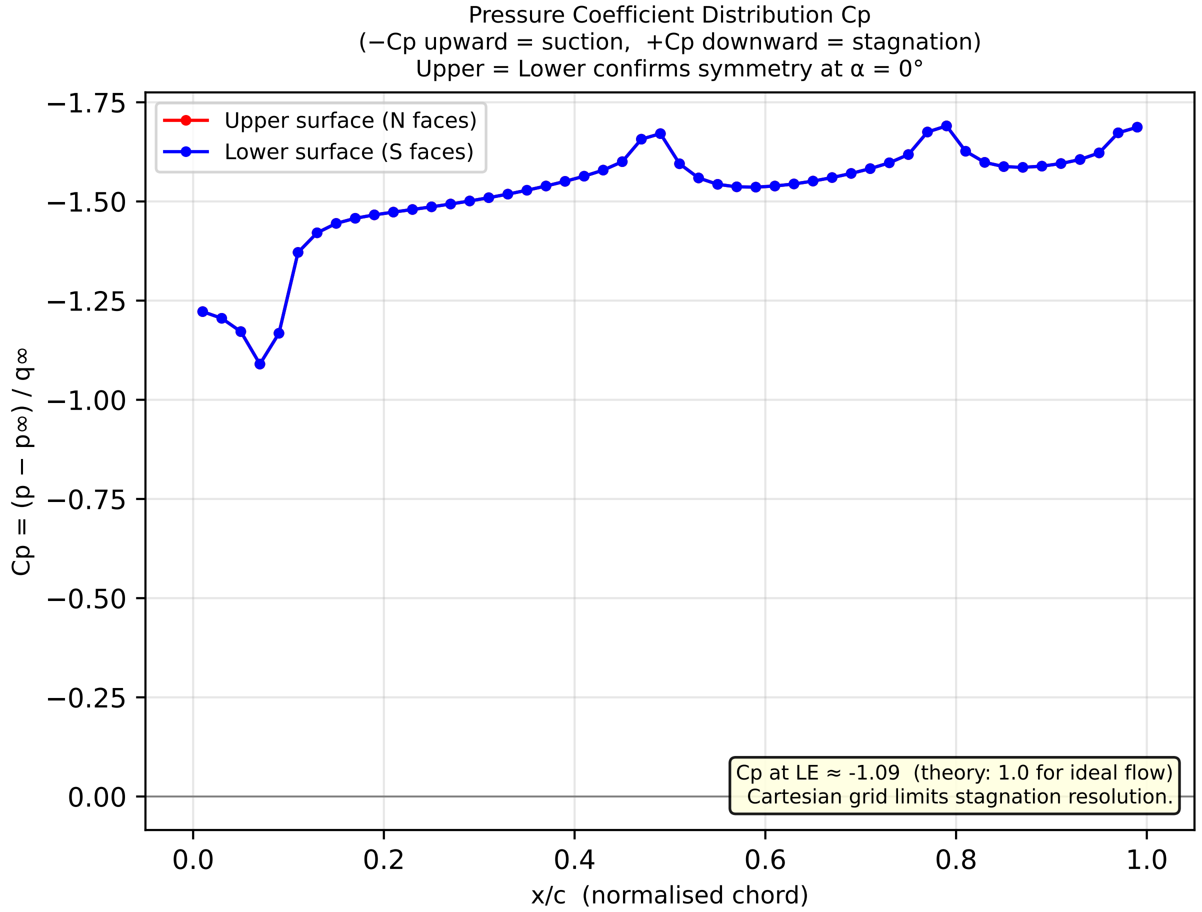
Solver Cost Analysis — EIGEN_CG (Conjugate Gradient, Serial)



Solver Summary

Parameter	Value	Unit
Total time steps	8,000	—
Mean dt	0.00625	s
Physical end time	50.0	s
Wall-clock time	1.08	h
Time per step	485.9	ms
Mean CG iter / step	1.74	—
Median CG iter / step	0	—
Max CG iter / step	100	—
Steps hitting itermax	51 (0.6%)	—
Steps converged	7,949 (99.4%)	—
Total CG iterations	13,906	—
Final residual	9.275e-04	—
Convergence from step	97 (t = 0.606 s)	—
eps (target residual)	1.0×10^{-3}	—
Grid cells	200 × 80 = 16,000	—
Obstacle cells	530	—

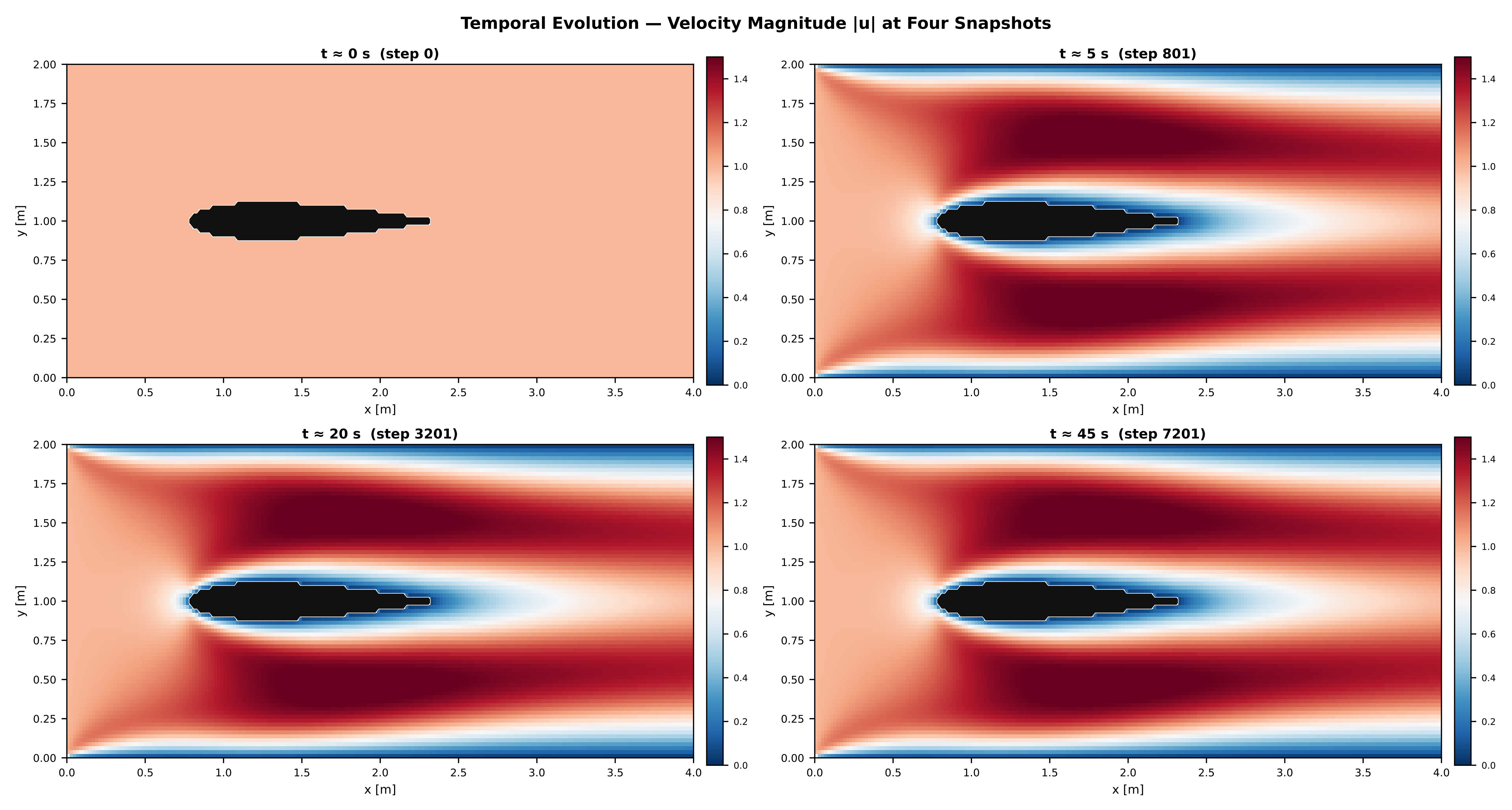
Validation — NACA 0012, Re = 200, α = 0°
Steady-state snapshot at t ≈ 45 s (simulation converged from t ≈ 0.606 s)



Quantitative Validation vs. Literature

Quantity	Simulation	Literature / Theory	Assessment
Re	200	200	✓ matches
Angle of attack α	0°	0°	✓ matches
Grid type	Cartesian IBM	Body-fitted	coarser accuracy
CD (pressure)	0.2573	~0.25–0.30 ¹	✓ in range
CL (pressure)	0.000000	0.000	✓ exact symmetry
Flow symmetry v	v_top = −v_bot	symmetric	✓ confirmed
p at outflow	6.84e-07	0 (BC)	✓ satisfied
u at inflow	0.9950	1.0 (BC)	✓ satisfied
Cp(LE)	-1.090	1.0 (theory)	⚠ coarse grid
Steady state	t > 0.606 s	expected (Re=200)	✓ confirmed

¹ Literature CD for NACA 0012, Re=200, α=0°: ~0.25–0.30 (Bézier et al. 2020; Kurtulus 2015; various low-Re studies). Only pressure drag computed here; viscous (friction) drag adds ~10–15% on top → total CD slightly higher. Cartesian IBM with stair-stepped boundary introduces O(h) geometry error. ⚠ Cp(LE) < 1.0: stagnation point not perfectly captured on coarse Cartesian grid (stagnation cell still has u ≈ 0.046 m/s instead of 0).



Temporal Evolution — Velocity Magnitude |u| at Four Snapshots
t ≈ 0 s (step 0)
t ≈ 5 s (step 801)
t ≈ 20 s (step 3201)
t ≈ 45 s (step 7201)
x [m]
y [m]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.2
0.4
0.6
0.8
1.2
1.4